



**Geospatial Competency Centre
Technology Services Division**

CoT Geocoder Technical Reference

Version 1.1

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Document Information

Revision History

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1 Section One – CoT Geocoder

1.1 Introduction

The RESTful services provide both an accurate and standardized method for validating and querying location data from the City's Geospatial Repository.

Enter an address, an intersection, or a place name within the City of Toronto and confirm its validity in addition to returning Administrative Areas. Additionally, a type-ahead "suggest" feature is required for the web interface.

1.2 Suggest Service

<https://map.toronto.ca/cotgeocoder/rest/geocoder/suggest>

Description: Search allows character-by-character autocomplete suggestions from user input with a minimum of 3 characters:

Required parameter:

searchString = [string] , user input string

Optional parameter:

- addressOnly = [integer] , enter '1' to return match address only or '0' to return matching address/intersection/place name (**deprecated**)
- matchAddress = [integer], enter '1' to match addresses or '0' to not match addresses. Default is '1'.
- matchIntersection = [integer], enter '1' to match intersections or '0' to not match intersections. Default is '1'.
- matchPlacename = [integer], enter '1' to match place names or '0' to not match place names. Default is '1'.
- callback = [string], callback parameter
- retRowLimit = [integer] , the maximum number of row return and the Default is 10

Output parameter:

- ADDRESS – match address/intersection/place name in text
- KEYSTRING – a key string "FEATURE_TYPE:geoid:[geoid]:rowid:[rowid]" can be used in the **findAddressCandidates** service for retrieving attributes of an address/intersection/a place name

NOTES:

This is an exact match search. It requires that the user input match an address, intersection, street name or place name as it is in the database. It should only be used for type ahead input controls and not fuzzy searches. There is no support for postal codes with suggest.

The following parameters should be used by Salesforce/311:

Return all feature types:

matchAddress=1

matchIntersection=1

matchPlacename=1

retRowLimit= (set higher than 10 if you need to see more)
searchString set to user's input as they type.

Example:
User types '55 Jo'

<https://map.toronto.ca/cotgeocoder/rest/geocoder/suggest?searchString=55%20jo&matchAddress=1&matchIntersection=1&matchStreetName=1&matchPlacename=1&f=pjson&retRowLimit=20>

It will return 14 matches returning Address and KeyString with the following format:

ADDRESS "55 Joanith Dr, East York"
KEYSTRING "ADDRESS:geoid:26708:rowid:387094"

To return full address information pass in the KeyString value to findAddressCandidates.

If no matches are found you must call findAddressCandidates and pass in the user input to the singleLine parameter. See findAddressCandidates for more information.

Sample call #1: User enters "55 j" as the search String, set maximum number of row return to 5, it returns a list of matching address

<https://map.toronto.ca/cotgeocoder/rest/geocoder/suggest?searchString=55%20j&matchAddress=1&matchIntersection=1&matchPlacename=1&f=pjson&retRowLimit=5>

```
{"result": {"rows": [  
  {  
    "ADDRESS": "55 Jackson Ave, Etobicoke",  
    "KEYSTRING": "ADDRESS:geoid:10079824:rowid:387075"  
  },  
  {  
    "ADDRESS": "55 Jacob Fisher Dr, Scarborough",  
    "KEYSTRING": "ADDRESS:geoid:348282:rowid:387076"  
  },  
  {  
    "ADDRESS": "55 James St, former Toronto",  
    "KEYSTRING": "ADDRESS:geoid:8087241:rowid:387077"  
  },  
  {  
    "ADDRESS": "55 James Foxway, North York",  
    "KEYSTRING": "ADDRESS:geoid:530820:rowid:387078"  
  },  
  {  
    "ADDRESS": "55 James Gray Dr, North York",  
    "KEYSTRING": "ADDRESS:geoid:9345399:rowid:387079"  
  }  
]}}
```

Note: System returns the first 5 addresses with number = 55 and street name starts with letter "j" in alphabetical order

Sample call #2: User enters "55 jo" as the search String

<https://map.toronto.ca/cotgeocoder/rest/geocoder/suggest?searchString=55%20jo&addressOnly=1&f=json>

```
{"result": {"rows": [
  {
    "ADDRESS": "55 Joanith Dr, East York",
    "KEYSTRING": "ADDRESS:geoid:26708:rowid:387094"
  },
  {
    "ADDRESS": "55 Joanna Dr, Scarborough",
    "KEYSTRING": "ADDRESS:geoid:348998:rowid:387095"
  },
  {
    "ADDRESS": "55 Jocelyn Cres, North York",
    "KEYSTRING": "ADDRESS:geoid:10145147:rowid:387096"
  },
  {
    "ADDRESS": "55 John St, former Toronto",
    "KEYSTRING": "ADDRESS:geoid:825934:rowid:387098"
  },
  {
    "ADDRESS": "55 John Lindsay Crt, North York",
    "KEYSTRING": "ADDRESS:geoid:2247472:rowid:387099"
  }
]}}
```

Note: System returns the first 10 addresses with number = 55 and street name starts with letter "jo" in alphabetical order

Sample call #3: User enters "fin". It returns suggestions of all types.

<https://map.toronto.ca/cotgeocoder/rest/geocoder/suggest?searchString=fin&f=json>

```
{"result": {"rows": [
  {
    "ADDRESS": "Finch and Adirondack, Scarborough",
    "KEYSTRING": "INTERSECTION:geoid:14412:rowid:549180"
  },
  {
    "ADDRESS": "Finch and Alamosa, North York",
    "KEYSTRING": "INTERSECTION:geoid:370320:rowid:549181"
  },
  {
    "ADDRESS": "Finch and Albion, Etobicoke",
    "KEYSTRING": "INTERSECTION:geoid:302630:rowid:549182"
  },
  {
    "ADDRESS": "Finch and Alexdon, North York",

```

```
    "KEYSTRING": "INTERSECTION:geoid:229738:rowid:549183"
  },
  {
    "ADDRESS": "Finch and Alexmuir, Scarborough",
    "KEYSTRING": "INTERSECTION:geoid:163785:rowid:549184"
  },
  {
    "ADDRESS": "Finch and Alness, North York",
    "KEYSTRING": "INTERSECTION:geoid:86407:rowid:549185"
  },
  {
    "ADDRESS": "Finch and Altamont, North York",
    "KEYSTRING": "INTERSECTION:geoid:26747:rowid:549186"
  },
  {
    "ADDRESS": "Finch and Ancona, North York",
    "KEYSTRING": "INTERSECTION:geoid:7018:rowid:549187"
  },
  {
    "ADDRESS": "Finch and Ardwick, North York",
    "KEYSTRING": "INTERSECTION:geoid:164874:rowid:549188"
  },
  {
    "ADDRESS": "Finch and Arrow, North York",
    "KEYSTRING": "INTERSECTION:geoid:210462:rowid:549189"
  }
]}}
```

Note: System returns the first 10 matching feature start with "fin". In this case, place name and intersection are returned.

Sample call #4: User enters "finch and y".

<https://map.toronto.ca/cotgeocoder/rest/geocoder/suggest?searchString=finch%20and%20y&f=pjson>

```
{"result": {"rows": [
  {
    "ADDRESS": "Finch and Yonge, North York",
    "KEYSTRING": "INTERSECTION:geoid:85922:rowid:549341"
  },
  {
    "ADDRESS": "Finch and York Gate, North York",
    "KEYSTRING": "INTERSECTION:geoid:261790:rowid:549342"
  },
  {
    "ADDRESS": "Finch Ave E and Yonge St, North York",
    "KEYSTRING": "INTERSECTION:geoid:85924:rowid:549419"
  },
  {
    "ADDRESS": "Finch Ave W and Yonge St, North York",
```



```
    "KEYSTRING": "INTERSECTION:geoid:85922:rowid:549511"
  },
  {
    "ADDRESS": "Finch Ave W and York Gate Blvd, North York",
    "KEYSTRING": "INTERSECTION:geoid:261790:rowid:549512"
  },
  {
    "ADDRESS": "Finch Corridor and York Gate, North York",
    "KEYSTRING": "INTERSECTION:geoid:427231:rowid:549556"
  },
  {
    "ADDRESS": "Finch Corridor and York University Busway, North York",
    "KEYSTRING": "INTERSECTION:geoid:426971:rowid:549557"
  },
  {
    "ADDRESS": "Finch Corridor Trl and York Gate Blvd, North York",
    "KEYSTRING": "INTERSECTION:geoid:427231:rowid:549602"
  },
  {
    "ADDRESS": "Finch Corridor Trl and York University Busway, North
York",
    "KEYSTRING": "INTERSECTION:geoid:426971:rowid:549603"
  }
]}}
```

Note: Always use "and" word in between the intersections

1.3 *findAddressCandidates* Service

<https://map.toronto.ca/cotgeocoder/rest/geocoder/findAddressCandidates>

Description:

- (1) Search by **keyString** returned from the **suggest** REST service.
- (2) Search by **singleLine** for input text string

Required parameter: **note: all parameters are case sensitive**

keyString = [string] , a key string in the format of "[FEATURE_TYPE]:geoid:[geoid]:rowid:[rowid]" and it is returned by the Suggest REST service

OR

singleLine = [string] , an input string can be full or part of an address, a place name, intersection separated by "and" or "/"

e.g. 100 queen, 100 queen w, city hall, m3v, Finch / Yong, Finch and Yonge

Optional parameters: **note: all parameters are case sensitive**

- unit = [string], unit to search for.
- outSrid = [integer] , location of X and Y in projection of 2019 for NAD27 MTM, 3857 for Web Mercator, and 4326. Default is set to 4326 WSG84.
- matchAddress = [integer], enter '1' to match addresses or '0' to not match addresses. Default is '1'.
- matchIntersection = [integer], enter '1' to match intersections or '0' to not match intersections. Default is '1'.
- matchPlacename = [integer], enter '1' to match place names or '0' to not match place names. Default is '1'.
- matchPostalCode = [integer], enter '1' to match postal codes or '0' to not match postal codes. Default is '1'.
- retRowLimit = [integer] , the maximum number of row return and default is 10

- areaTypeCode1 = [string], enter area type code to return corresponding administrative area information. For example: enter "CITW" to return electoral ward attribute.
- areaTypeCode2 = [string], enter area type code to return corresponding administrative area information.
- areaTypeCode3 = [string], enter area type code to return corresponding administrative area information.
- areaTypeCode4 = [string], enter area type code to return corresponding administrative area information.
- areaTypeCode5 = [string], enter area type code to return corresponding administrative area information.

Commonly referenced area type codes are listed below:

areaTypeCode	Description
MUN	Former Municipality
CITW	Toronto City Ward
TWSA	Toronto Water Service Area
NEIG	Neighborhood
CENS	Census Tract
CNBH	Census Neighborhood
FORD	Forestry District
IBMD	IBMS District
IBMG	IBMS Grid
TIA	TMMS investigator area

* Refer to Appendix for the full List of Area Types

NOTES:

If no results are found in suggest service, you should pass the user input into findAddressCandidates into the singleLine parameter to do a fuzzy search. No postal code searches are supported by suggest so you will need to pass postal code into this service. This can be done in the front end interface by calling findAddressCandidates when the user hits the 'enter' key. When user is typing and suggest is being used, if user selects a return from suggest, pass in the KeyString to findAddressCandidates. If there is no selection and user hits the 'enter' key call findAddressCandidate to pass input to singleLine. This method also supports Soundex.

The following parameters should be used by Salesforce/311:

Return area type codes for municipality, ward, and Toronto Water service area:

- areaTypeCode1= MUN
- areaTypeCode2= CITW
- areaTypeCode3=TWSA

Return all feature types:

- matchAddress=1
- matchIntersection=1
- matchPlacename=1
- matchPostalCode=1

Return all units:

- unit=%

1.3.1 findAddressCandidates by keyString

Search by **keyString** returned from the Suggest REST service.

The result in below was returned by the Suggest REST service call

<https://map.toronto.ca/cotgeocoder/rest/geocoder/suggest?searchString=55%20john%20st&addressOnly=1&f=json>

```
{"result": {"rows": [
  {
    "ADDRESS": "55 John St, York",
    "KEYSTRING": "ADDRESS:geoid:11297211:rowid:387097"
  },
  {
    "ADDRESS": "55 John St, former Toronto",
    "KEYSTRING": "ADDRESS:geoid:825934:rowid:387098"
  },
  {
    "ADDRESS": "55 John Stoner Dr, Scarborough",
    "KEYSTRING": "ADDRESS:geoid:349059:rowid:387100"
  }
]}}
```

Sample Response #1:

Using the **keyString** = ADDRESS:geoid:825934:rowid:387098, the detail attributes of 55 John St, former Toronto can be retrieved.

<https://map.toronto.ca/cotgeocoder/rest/geocoder/findAddressCandidates?f=json&keyString=ADDRESS:geoid:825934:rowid:387098>

```
{
  "result": {
    "rows": {
      "MUNICIPALITY": "former Toronto",
      "MATCHTYPE": "MATCH",
      "PLACE_NAME": "Metro Hall",
      "MAINT_STAGE": "REGULAR",
      "LO_NUM": "55",
      "KEY_DESC": "55 John St, former Toronto",
      "LINEAR_NAME": "John",
      "GENERAL_USE_CODE": "",
      "LATITUDE": "43.646162378",
      "GTYPE": "Point",
      "OUTY": "43.646162378",
      "OUTX": "-79.3889347303",
      "LINEAR_NAME_TYPE": "St",
      "ADDP_GEO_ID": "825934",
      "HI_NUM": "",
      "DETAIL": "55 John St, M5V, former Toronto, Metro Hall, Ward: Spadina-Fort York",
      "WARD_NUMBER": "10",
      "Y": "4833807.374",
      "X": "313744.473",
      "LINEAR_NAME_DIR": "",
      "GENERAL_USE": ""
    }
  }
}
```

```
    "CENTRELINE_OFFSET": "51.03",
    "ADDRESS_CLASS": "L",
    "ADDRESS_FULL": "55 John St",
    "CENTRELINE_SIDE": "R",
    "ADDRESS_CLASS_DESC": "Land",
    "CENTRELINE_ID": "1146312",
    "SCORE": "100",
    "WARD": "Spadina-Fort York",
    "FEATURECLASS": "Address",
    "LINEAR_NAME_ID": "9747",
    "NUMBER_OF_UNITS": "0",
    "LO_NUM_SUF": "",
    "ADDRESS_STATUS": "",
    "HI_NUM_SUF": "",
    "ADDRESS_POINT_ID_LINK": "",
    "CITY": "Toronto",
    "LONGITUDE": "-79.3889347303",
    "POSTAL_CODE": "M5V",
    "LINEAR_NAME_DESC": "TORONTO",
    "CENTRELINE_MEASURE": "86.9"
  }
}
```

Note: No outSrid has been given, the OUTX and OUTY are in WGS84 by default

Sample Response #2

<https://map.toronto.ca/cotgeocoder/rest/geocoder/findAddressCandidates?f=jsonkeyString=ADDRESS:geoid:825934:rowid:387098&areaTypeCode1=CITW&areaTypeCode2=NEIG>

```
{
  "result": {
    "rows": {
      "MUNICIPALITY": "former Toronto",
      "MATCHTYPE": "MATCH",
      "PLACE_NAME": "Metro Hall",
      "MAINT_STAGE": "REGULAR",
      "LO_NUM": "55",
      "KEY_DESC": "55 John St, former Toronto",
      "LINEAR_NAME": "John",
      "GENERAL_USE_CODE": "",
      "LATITUDE": "43.646162378",
      "GTYPE": "Point",
      "OUTY": "43.646162378",
      "OUTX": "-79.3889347303",
      "LINEAR_NAME_TYPE": "St",
      "ADDP_GEO_ID": "825934",
      "HI_NUM": "",
      "DETAIL": "55 John St, M5V, former Toronto, Metro Hall, Ward: Spadina-Fort York",
      "WARD_NUMBER": "10",
    }
  }
}
```

```
"Y": "4833807.374",
"X": "313744.473",
"LINEAR_NAME_DIR": "",
"GENERAL_USE": "",
"CENTRELINE_OFFSET": "51.03",
"ADDRESS_CLASS": "L",
"ADDRESS_FULL": "55 John St",
"CENTRELINE_SIDE": "R",
"ADDRESS_CLASS_DESC": "Land",
"CENTRELINE_ID": "1146312",
"SCORE": "100",
"WARD": "Spadina-Fort York",
"FEATURECLASS": "Address",
"LINEAR_NAME_ID": "9747",
"NUMBER_OF_UNITS": "0",
"AREACURS2": {
  "array": {
    "AREA_NAME": "04000000",
    "AREA_DESC": "04000000",
    "TYPE_DESCRIPTION": "Neighborhood",
    "AREA_SHORT_CODE": "000",
    "AREA_ID": "2533179",
    "AREA_LONG_CODE": "04000000",
    "AREA_TYPE": "NEIG"
  }
},
"AREACURS1": {
  "array": {
    "AREA_NAME": "Spadina-Fort York",
    "AREA_DESC": "Spadina-Fort York (10)",
    "TYPE_DESCRIPTION": "Toronto Municipal Electoral Ward",
    "AREA_SHORT_CODE": "10",
    "AREA_ID": "2702250",
    "AREA_LONG_CODE": "10",
    "AREA_TYPE": "CITW"
  }
},
"LO_NUM_SUF": "",
"ADDRESS_STATUS": "",
"HI_NUM_SUF": "",
"ADDRESS_POINT_ID_LINK": "",
"CITY": "Toronto",
"LONGITUDE": "-79.3889347303",
"POSTAL_CODE": "M5V",
"LINEAR_NAME_DESC": "TORONTO",
"CENTRELINE_MEASURE": "86.9"
}
}
```

Note: The same address is returned with its corresponding Ward and Neighborhood.

1.3.2 findAddressCandidates by singleLine text

Sample Response #1: search by input string of "18 dyas"

<https://map.toronto.ca/cotgeocoder/rest/geocoder/findAddressCandidates?f=json&singleLine=18%20dyas>

```
{
  "result": {
    "rows": {
      "MUNICIPALITY": "North York",
      "MATCHTYPE": "CANDIDATE",
      "PLACE_NAME": "",
      "MAINT_STAGE": "REGULAR",
      "LO_NUM": "18",
      "KEY_DESC": "18 Dyas Rd (Total 7 Units)",
      "LINEAR_NAME": "Dyas",
      "GENERAL_USE_CODE": "",
      "LATITUDE": "43.751706789",
      "GTYPE": "Point",
      "OUTY": "43.751706789",
      "OUTX": "-79.352625219",
      "LINEAR_NAME_TYPE": "Rd",
      "ADDP_GEO_ID": "13967566",
      "HI_NUM": "",
      "DETAIL": "18 Dyas Rd, M3B, North York, Ward: Don Valley East",
      "WARD_NUMBER": "16",
      "Y": "4845538.389",
      "X": "316652.641",
      "LINEAR_NAME_DIR": "",
      "GENERAL_USE": "",
      "CENTRELINE_OFFSET": "71.01",
      "ADDRESS_CLASS": "L",
      "ADDRESS_FULL": "18 Dyas Rd",
      "CENTRELINE_SIDE": "R",
      "ADDRESS_CLASS_DESC": "Land",
      "CENTRELINE_ID": "11541978",
      "SCORE": "97",
      "WARD": "Don Valley East",
      "FEATURECLASS": "Address",
      "LINEAR_NAME_ID": "5467",
      "NUMBER_OF_UNITS": "7",
      "LO_NUM_SUF": "",
      "ADDRESS_STATUS": "",
      "HI_NUM_SUF": "",
      "ADDRESS_POINT_ID_LINK": "",
      "CITY": "Toronto",
      "LONGITUDE": "-79.352625219",
      "POSTAL_CODE": "M3B",
      "LINEAR_NAME_DESC": "",
      "CENTRELINE_MEASURE": "228.94"
    }
  }
}
```

```
}  
}
```

Note: No outSrid has been given, the OUTX and OUTY are in WGS84 by default

Sample Response #2: search by input string of "18 dyas", outSrid is set to 3857 (Web Mercator) and areaTypeCode1 = 'AD' (Assessment Division)

<https://map.toronto.ca/cotgeocoder/rest/geocoder/findAddressCandidates?f=json&singleLine=18%20dyas&outSrid=3857&areaTypeCode1=AD>

```
{  
  "result": {  
    "rows": {  
      "MUNICIPALITY": "North York",  
      "MATCHTYPE": "CANDIDATE",  
      "PLACE_NAME": "",  
      "MAINT_STAGE": "REGULAR",  
      "LO_NUM": "18",  
      "KEY_DESC": "18 Dyas Rd (Total 7 Units)",  
      "LINEAR_NAME": "Dyas",  
      "GENERAL_USE_CODE": "",  
      "LATITUDE": "43.751706789",  
      "GTYPE": "Point",  
      "OUTY": "5427098.2929748",  
      "OUTX": "-8833493.8324886",  
      "LINEAR_NAME_TYPE": "Rd",  
      "ADDP_GEO_ID": "13967566",  
      "HI_NUM": "",  
      "DETAIL": "18 Dyas Rd, M3B, North York, Ward: Don Valley East",  
      "WARD_NUMBER": "16",  
      "Y": "4845538.389",  
      "X": "316652.641",  
      "LINEAR_NAME_DIR": "",  
      "GENERAL_USE": "",  
      "CENTRELINE_OFFSET": "71.01",  
      "ADDRESS_CLASS": "L",  
      "ADDRESS_FULL": "18 Dyas Rd",  
      "CENTRELINE_SIDE": "R",  
      "ADDRESS_CLASS_DESC": "Land",  
      "CENTRELINE_ID": "11541978",  
      "SCORE": "97",  
      "WARD": "Don Valley East",  
      "FEATURECLASS": "Address",  
      "LINEAR_NAME_ID": "5467",  
      "NUMBER_OF_UNITS": "7",  
      "AREACURSOR1": {  
        "array": {  
          "AREA_NAME": "08102",  
          "AREA_DESC": "08102",  
          "TYPE_DESCRIPTION": "Assessment Division",  

```



```
        "AREA_SHORT_CODE": "2",
        "AREA_ID": "457271",
        "AREA_LONG_CODE": "08102",
        "AREA_TYPE": "AD"
    },
    "LO_NUM_SUF": "",
    "ADDRESS_STATUS": "",
    "HI_NUM_SUF": "",
    "ADDRESS_POINT_ID_LINK": "",
    "CITY": "Toronto",
    "LONGITUDE": "-79.352625219",
    "POSTAL_CODE": "M3B",
    "LINEAR_NAME_DESC": "",
    "CENTRELINE_MEASURE": "228.94"
}
}
```

Note that the OUTX and OUTY are in Web Mercator and the corresponding assessment division information is shown in Blue.

Sample Response #3: search for intersection by an input string "finch / yong"

<https://map.toronto.ca/cotgeocoder/rest/geocoder/findAddressCandidates?f=json&singleLine=finch%20/%20yong>

```
{
  "result": {
    "rows": [
      {
        "MATCHTYPE": "CANDIDATE",
        "LINEAR_NAME_TYPE_TO": "St",
        "KEY_DESC": "Finch Ave E \\/ Yonge St",
        "GTYPE": "Point",
        "LATITUDE": "43.779769082",
        "OUTY": "43.779769082",
        "OUTX": "-79.415577699",
        "TURN_GEO_ID": "85924",
        "LINEAR_NAME_FULL_FROM": "Finch Ave E",
        "LINEAR_NAME_TO": "Yonge",
        "LINEAR_NAME_FULL_TO": "Yonge St",
        "DETAIL": "Finch Ave E \\/ Yonge St \\/ Finch Ave W",
        "CLASSIFICATION": "MJRSL",
        "INT_GEO_ID": "13446142",
        "Y": "4848648.953",
        "X": "311579.857",
        "LINEAR_NAME_DIR_FROM": "E",
        "LINEAR_NAME_ID_FROM": "5589",
        "CLASSIFICATION_DESC": "Major-Single Level",
        "LINEAR_NAME_ID_TO": "4733",
        "LINEAR_NAME_DESC_FROM": ""
      }
    ]
  }
}
```

```
"LINEAR_NAME_TYPE_FROM": "Ave",
"SCORE": "84",
"LINEAR_NAME_FROM": "Finch",
"FEATURECLASS": "Intersection",
"LINEAR_NAME_DIR_TO": "",
"LINEAR_NAME_DESC_TO": "",
"LONGITUDE": "-79.415577699"
},
{
  "MATCHTYPE": "CANDIDATE",
  "LINEAR_NAME_TYPE_TO": "St",
  "KEY_DESC": "Finch Ave W \\/ Yonge St",
  "GTYPE": "Point",
  "LATITUDE": "43.779769082",
  "OUTY": "43.779769082",
  "OUTX": "-79.415577699",
  "TURN_GEO_ID": "85922",
  "LINEAR_NAME_FULL_FROM": "Finch Ave W",
  "LINEAR_NAME_TO": "Yonge",
  "LINEAR_NAME_FULL_TO": "Yonge St",
  "DETAIL": "Finch Ave W \\/ Yonge St \\/ Finch Ave E",
  "CLASSIFICATION": "MJRSL",
  "INT_GEO_ID": "13446142",
  "Y": "4848648.953",
  "X": "311579.857",
  "LINEAR_NAME_DIR_FROM": "W",
  "LINEAR_NAME_ID_FROM": "1610",
  "CLASSIFICATION_DESC": "Major-Single Level",
  "LINEAR_NAME_ID_TO": "4733",
  "LINEAR_NAME_DESC_FROM": "",
  "LINEAR_NAME_TYPE_FROM": "Ave",
  "SCORE": "84",
  "LINEAR_NAME_FROM": "Finch",
  "FEATURECLASS": "Intersection",
  "LINEAR_NAME_DIR_TO": "",
  "LINEAR_NAME_DESC_TO": "",
  "LONGITUDE": "-79.415577699"
}
]
}
```

Note: Alternate input string is "finch and yong"

1.4 reverseGeocode Service

<https://map.toronto.ca/cotgeocoder/rest/geocoder/reverseGeocode>

Description: Find the nearest address or intersection by coordinates

Required parameter:

- x = [float] , x co-ordinate or longitude
- y = [float] y co-ordinate or latitude

Optional parameter:

- inSrid = [integer], enter SRID for input co-ordinate. Default is '4326'.
- outSrid = [integer], enter SRID for output co-ordinate. Default is '4326'
- searchDistance = [integer], enter max search distance(m). Must be less than 1000.
- matchAddress = [integer], enter '1' to match addresses or '0' to not match addresses. Default is '1'.
- matchIntersection = [integer], enter '1' to match intersections or '0' to not match intersections. Default is '1'.
- callback = [string], callback parameter
- retRowLimit = [integer] , the maximum number of row return and the Default is 10
- areaTypeCode1 = [string], enter area type code to return corresponding administrative area information. For example: enter "CITW" to return electoral ward attributes
- areaTypeCode2 = [string], enter area type code to return corresponding administrative area information.
- areaTypeCode3 = [string], enter area type code to return corresponding administrative area information.
- areaTypeCode4 = [string], enter area type code to return corresponding administrative area information.
- areaTypeCode5 = [string], enter area type code to return corresponding administrative area information.
- f = [string], format of return, either json, xml or pjson.

NOTES:

The following parameters should be used by Salesforce/311:

Return area type codes for municipality, ward, and Toronto Water service area:

- areaTypeCode1= MUN
- areaTypeCode2= CITW
- areaTypeCode3=TWSA

Return all feature types:

- matchAddress=1
- matchIntersection=1

Set inSrid and outSrid:

- inSrid = 4326
- outSrid = 4326

Set the x,y co-ordinates.

Sample Response #1: search for address from a given co-ordinate

<https://map.toronto.ca/cotgeocoder/rest/geocoder/reverseGeocode?x=->

[79.51587&y=43.7033&inSrid=4326&outSrid=4326&areaTypeCode1=MUN&areaTypeCode2=CITW&areaTypeCode3=TWSA&f=json&matchAddress=1&matchIntersection=0](https://api.toronto.ca/geocoders/v1.1/79.51587&y=43.7033&inSrid=4326&outSrid=4326&areaTypeCode1=MUN&areaTypeCode2=CITW&areaTypeCode3=TWSA&f=json&matchAddress=1&matchIntersection=0)

```
{"result": {"rows": [{
  "MUNICIPALITY": "York",
  "MATCHTYPE": "CANDIDATE",
  "PLACE_NAME": null,
  "MAINT_STAGE": "REGULAR",
  "LO_NUM": 57,
  "KEY_DESC": "57 John St, York (Proximity:11.19m)",
  "LINEAR_NAME": "John",
  "GENERAL_USE_CODE": null,
  "LATITUDE": 43.7033921433,
  "GTYPE": "Point",
  "OUTY": 43.7033921433,
  "OUTX": -79.5158117254,
  "LINEAR_NAME_TYPE": "St",
  "ADDP_GEO_ID": 11297213,
  "HI_NUM": null,
  "DETAIL": "57 John St, M9N,York, Ward: York South-Weston (11)",
  "WARD_NUMBER": "11",
  "Y": 4840159.419,
  "X": 303510.083,
  "LINEAR_NAME_DIR": null,
  "GENERAL_USE": null,
  "CENTRELINE_OFFSET": 17.87,
  "ADDRESS_CLASS": "L",
  "ADDRESS_FULL": "57 John St",
  "CENTRELINE_SIDE": "R",
  "ADDRESS_CLASS_DESC": "Land",
  "CENTRELINE_ID": 11297207,
  "SCORE": 99,
  "WARD": "York South-Weston (11)",
  "FEATURECLASS": "Address",
  "AREACURSOR3": {"array": [
    {
      "AREA_NAME": "WEST",
      "AREA_DESC": "WEST",
      "TYPE_DESCRIPTION": "Toronto Water Service Area",
      "AREA_SHORT_CODE": "WE",
      "AREA_ID": 2792868,
      "AREA_LONG_CODE": "WE",
      "AREA_TYPE": "TWSA"
    }
  ]},
  "LINEAR_NAME_ID": 9831,
  "NUMBER_OF_UNITS": 0,
  "AREACURSOR2": {"array": [
    {
      "AREA_NAME": null,
      "AREA_DESC": "York South-Weston (11)",
      "TYPE_DESCRIPTION": "Toronto Municipal Electoral Ward",
      "AREA_SHORT_CODE": "11",
      "AREA_ID": 14630017,
      "AREA_LONG_CODE": "WE11",
    }
  ]}
}]}
```

```
"AREA_TYPE": "CITW"
}}},
"AREACURS01": {"array": [  {
  "AREA_NAME": null,
  "AREA_DESC": "YORK",
  "TYPE_DESCRIPTION": "Former Local Municipality",
  "AREA_SHORT_CODE": "14",
  "AREA_ID": 49884,
  "AREA_LONG_CODE": "14",
  "AREA_TYPE": "MUN"
}}},
"LO_NUM_SUF": null,
"ADDRESS_STATUS": null,
"HI_NUM_SUF": null,
"ADDRESS_POINT_ID_LINK": null,
"CITY": "Toronto",
"LONGITUDE": -79.5158117254,
"POSTAL_CODE": "M9N",
"LINEAR_NAME_DESC": "YORK",
"CENTRELINE_MEASURE": 85.32
}}}]}
```

Sample Response #2: search for intersection from a given co-ordinate

<https://map.toronto.ca/cotgeocoder/rest/geocoder/reverseGeocode?x=-79.51587&y=43.7033&inSrid=4326&outSrid=4326&areaTypeCode1=MUN&areaTypeCode2=CITW&areaTypeCode3=TWSA&f=json&matchAddress=0&matchIntersection=1>

```
{"result": {"rows": [{
  "MATCHTYPE": "CANDIDATE",
  "LINEAR_NAME_TYPE_TO": "Ave",
  "KEY_DESC": "John St / Rosemount Ave (Proximity:77.24m)",
  "GTYPE": "Point",
  "LATITUDE": 43.7028510062,
  "OUTY": 43.7028510062,
  "OUTX": -79.5166002344,
  "TURN_GEO_ID": 40632,
  "LINEAR_NAME_FULL_FROM": "John St",
  "LINEAR_NAME_TO": "Rosemount",
  "LINEAR_NAME_FULL_TO": "Rosemount Ave",
  "DETAIL": "John St / Rosemount Ave",
  "CLASSIFICATION": "MNRSL",
  "INT_GEO_ID": 13457994,
  "Y": 4840099.317,
  "X": 303446.518,
  "LINEAR_NAME_DIR_FROM": null,
  "LINEAR_NAME_ID_FROM": 9831,
  "CLASSIFICATION_DESC": "Minor-Single Level",
  "LINEAR_NAME_ID_TO": 855,
  "LINEAR_NAME_DESC_FROM": "YK",
  "LINEAR_NAME_TYPE_FROM": "St",
```

```
"SCORE": 99,
"LINEAR_NAME_FROM": "John",
"FEATURECLASS": "Intersection",
"AREACURSOR3": {"array": [  {
  "AREA_NAME": "WEST",
  "AREA_DESC": "WEST",
  "TYPE_DESCRIPTION": "Toronto Water Service Area",
  "AREA_SHORT_CODE": "WE",
  "AREA_ID": 2792868,
  "AREA_LONG_CODE": "WE",
  "AREA_TYPE": "TWSA"
}]],
"AREACURSOR4": {"array": [  {
  "AREA_NAME": "WEST",
  "AREA_DESC": "WEST",
  "TYPE_DESCRIPTION": "Toronto Water Service Area",
  "AREA_SHORT_CODE": "WE",
  "AREA_ID": 2792868,
  "AREA_LONG_CODE": "WE",
  "AREA_TYPE": "TWSA"
}]],
"AREACURSOR2": {"array": [  {
  "AREA_NAME": null,
  "AREA_DESC": "York South-Weston (11)",
  "TYPE_DESCRIPTION": "Toronto Municipal Electoral Ward",
  "AREA_SHORT_CODE": "11",
  "AREA_ID": 14630017,
  "AREA_LONG_CODE": "WE11",
  "AREA_TYPE": "CITW"
}]],
"AREACURSOR1": {"array": [  {
  "AREA_NAME": "York",
  "AREA_DESC": "York",
  "TYPE_DESCRIPTION": "Former Local Municipality",
  "AREA_SHORT_CODE": "14",
  "AREA_ID": 49884,
  "AREA_LONG_CODE": "14",
  "AREA_TYPE": "MUN"
}]],
"LINEAR_NAME_DIR_TO": null,
"LINEAR_NAME_DESC_TO": "NY, YK",
"LONGITUDE": -79.5166002344
}}}}
```

2 Section Two – CoT Geocoder (esri format)

Notes: to be deleted

- (1) Need to return geo_id for Single Line Input search
- (2) ReverseGeocode does not work outside

2.1 Introduction

The RESTful service provides location search and validation services using the esri method. It is based on [esri ArcGIS World Geocoding Service](#), implemented with selective operations and parameters. The key difference from the original esri implementation is that the data is directly sourced from the City's Geospatial Data Repository while the location of interest is within the geographic boundary of the City of Toronto.

Operations implemented:

- suggest
- findAddressCandidates
- reverseGeocode

2.2 Suggest Service (esri format):

Suggest service allows character-by-character autocomplete suggestions from user input with a minimum of 3 characters. This search works for locations within or outside of Toronto. This service returns the following types of locations:

- Addresses (i.e. 55 Yonge St, former Toronto)
- Intersections (i.e. Bloor and Bathurst, Toronto)
- Place names (i.e. Nathan Phillips Square, 100 Queen St W, former Toronto)

<https://map.toronto.ca/geoservices/rest/services/gccgeocoder/GeocodeServer/suggest>

Input Parameters

Parameter	Type	Required	Description
text	string	Yes	Search string Minimum of 3 chars (including the space)
torontoOnly	Boolean	Yes	Toronto Only < true false > Default is false
useWorldCoverage	Boolean	No	Use world coverage to search
maxSuggestions	integer	No	the maximum number of rows returned Default is 20
f	string	No	Format <json pjson xml> Default is json

Output Parameters

Parameter	Type	Description
magicKey	string	An unique transient identifier assigned to a matched location as in text , e.g, gcc52dba7b-29e8-4517-9540-90ff76b7c1bcgeoid390002typeADDRESS for 55 John St, former Toronto It can be used in the findAddressCandidates service for retrieving attributes of address or intersection
isCollection	Boolean	Is Collection < true false >
text	string	Address, Intersection or Place name that started with the text 55 John St, former Toronto

Search addresses that start with "55 john" within or outside Toronto

- text = 55 john
- torontoOnly = false
- f = pjson

<https://map.toronto.ca/geoservices/rest/services/gccgeocoder/GeocodeServer/suggest?text=55%20john&torontoOnly=false&f=pjson>

Result

```
{
  "suggestions": [
    {
      "isCollection": false,
      "magicKey": "gcc89005920-8a01-46a9-933c-a9043f6434c4geoid391606typeADDRESS",
      "text": "55 John Lindsay Crt, North York"
    },
    {
      "isCollection": false,
      "magicKey": "gccac318b1c-9319-4995-a485-687959bbab35geoid391607typeADDRESS",
      "text": "55 John Perkins Bull Dr, North York"
    },
    {
      "isCollection": false,
      "magicKey": "gcc4a5f853f-32f6-4672-9c1c-6e6ca6366f51geoid391604typeADDRESS",
      "text": "55 John St, former Toronto"
    },
    {
      "isCollection": false,
      "magicKey": "gcc8ee521f-5f80-4375-a387-5c050601009dgeoid391605typeADDRESS",
      "text": "55 John St, York"
    },
    {
      "isCollection": false,
      "magicKey": "gcc108ba4b1-1c84-499a-baec-8b231b004154geoid391608typeADDRESS",
      "text": "55 John Stoner Dr, Scarborough"
    }
  ]
}
```

```
{,
{
  "isCollection": false,
  "magicKey": "gcc61e6ec92-da35-48af-8b0a-6c4fced079bageoid391609typeADDRESS",
  "text": "55 John Tabor Trl, Scarborough"
},
{
  "isCollection": false,
  "magicKey":
  "dHA9MCNsb2M9ODk2NDQ0NiNsbmc9MzMjaG49NTUjbGJzPTEwOT00MDQyNTk2Mw==",
  "text": "55 John St W, Niagara-on-the-Lake, Ontario, L0S, CAN"
},
{
  "isCollection": false,
  "magicKey":
  "dHA9MCNsb2M9ODk2NDcyNiNsbmc9MzMjaG49NTUjbGJzPTEwOT00MDQyNTk1Mw==",
  "text": "55 John St E, Niagara-on-the-Lake, Ontario, L0S, CAN"
},
{
  "isCollection": false
  "magicKey":
  "dHA9MCNsb2M9ODk2NDY1MCNsbmc9MzMjaG49NTUjbGJzPTEwOT00MDQyNTQ0Ng==",
  "text": "55 Johnson St, Niagara-on-the-Lake, Ontario, L0S 1J0, CAN"
},
{
  "isCollection": false,
  "magicKey":
  "dHA9MCNsb2M9OTA3NDEzMjNsbmc9MzMjaG49NTUjbGJzPTEwOT00MDQyNTk1OQ==",
  "text": "55 John St S, Mississauga, Ontario, L5H, CAN"
}
}]}
```

NOTES:

This is an exact match search. It requires that the user input match an address, intersection, street name or place name as it is in the database. It should only be used for type ahead input controls and not fuzzy searches. There is no support for postal codes with suggest.

2.3 findAddressCandidates Service (esri format)

<https://map.toronto.ca/geoservices/rest/services/gccgeocoder/GeocodeServer/findAddressCandidates>

Description:

(3) Search by **magicKey** returned from the **suggest** REST service.

(4) Search by **Single Line Input** for input text string

Note: all parameters are case sensitive

Input Parameters

Parameter	Type	Required	Description
Single Line Input	string	Yes	an input string can be full or part of an address, a place name, intersection separated by "and" or "/" e.g. 100 queen, 100 queen w, city hall, m3v, Finch / Yong, Finch and Yonge discarded if magicKey is used
outSR	string	No	Output Spatial Reference ID Location of X/Longitude and Y/Latitude in coordinate system of: * 2019 – NAD27 MTM * 3857 – Web Mercator * 4326 – WSG84 * 2952 – NAD83 CSRS Default is WSG84 4326
torontoOnly	boolean	No	Toronto Only < true false > Default is false
useWorldCoverage	boolean	No	Use world coverage for search
outFields	integer	No	Out Fields – fields to return * for all fields
magicKey	integer	No	a key string is formatted as - Inside Toronto, gcc[random string]geoid[geoid]type[feature type], e.g. "gcc89005920-8a01-46a9-933c-a9043f6434c4geoid391606typeADDRESS" - Outside of Toronto, use esri World Geocoder magicKey, e.g. "dHA9MCNsb2M9ODk2NDcyNiNsbmc9MzMjaG49NTUjbGJzPTEwOTo0MDQyNTk1Mw=="
maxLocations	integer	No	Maximum number of rows to return Default is 20
f	string	No	Format <json pjson xml> Default is json

NOTES:

If no results are found in suggest service, the user input maybe passed to findAddressCandidates into the **single Line Input** parameter to do a fuzzy search. This can be done in the front end interface by calling **findAddressCandidates** when the user hits the 'enter' key. When user is typing and suggest is being used, if user selects a return from suggest, pass in the **magicKey** to findAddressCandidates. If there is no selection and user hits the 'enter' key call **findAddressCandidate** to pass the input to **single Line Input**. This method also supports Soundex.

2.3.1 findAddressCandidates by magicKey

Search by **magicKey** returned from the Suggest REST service.

The result in below was returned by the Suggest REST service call

<https://map.toronto.ca/geoservices/rest/services/gccgeocoder/GeocodeServer/suggest?text=18%20dyas&torontoOnly=false&f=pjson>

```
{"suggestions": [
  {
    "isCollection": false,
    "magicKey": "gcc5bd46332-69d7-4e69-b53b-145b6a8a2afcgeoid129457typeADDRESS",
    "text": "18 Dyas Rd, North York"
  },
  {
    "isCollection": false,
    "magicKey": "dHA9MCNsb2M9MTM1ODQ4MzMjbG5nPTMzI2huPTE4I2xicz0xMDk6NDUyNDU2MDM=",
    "text": "18 Dyana Dr, Lindsay, Kawartha Lakes, Ontario, K9V 5A3, CAN"
  },
  {
    "isCollection": false,
    "magicKey": "dHA9MCNsb2M9MTM1ODQ4MzMjbG5nPTMzI2xicz0xMDk6NDUyNDU2MDM=",
    "text": "Dyana Dr, Lindsay, Kawartha Lakes, Ontario, K9V, CAN"
  }
]}
```

Sample Response #1:

Using the magicKey = gcc5bd46332-69d7-4e69-b53b-145b6a8a2afcgeoid129457typeADDRESS, the detail attributes of 18 Dyas Rd, North York can be retrieved.

https://map.toronto.ca/geoservices/rest/geocodeserver/findAddressCandidates?Single%20Line%20Input=&f=pjson&outSR=%7B%22wkid%22%3A4326%7D&outFields=*&magicKey=gcc5bd46332-69d7-4e69-b53b-145b6a8a2afcgeoid129457typeADDRESS&maxLocations=6

```
{
  "candidates": [{
    "address": "18 Dyas Rd, North York, Ontario, M3B",
```

```
"attributes": {
  "AddBldg": "",
  "AddNum": "18",
  "AddNumFrom": "ADDNUMFROM",
  "AddNumTo": "ADDNUMTO",
  "Addr_type": "PointAddress",
  "City": "North York",
  "Country": "CAN",
  "DisplayX": -79.352625219,
  "DisplayY": 43.751706789,
  "Distance": 0,
  "LangCode": "ENG",
  "Loc_name": "CAN.PointAddress",
  "Match_addr": "18 Dyas Rd, North York",
  "Nbrhd": "",
  "Phone": "",
  "PlaceName": "",
  "Place_addr": "",
  "Postal": "M3B",
  "PostalExt": "",
  "Rank": "",
  "Region": "Ontario",
  "Score": 80,
  "Side": "R",
  "StAddr": "18 Dyas Rd",
  "StDir": "",
  "StName": "Dyas",
  "StPreDir": "",
  "StPreType": "",
  "StType": "Rd",
  "Subregion": "",
  "Type": "",
  "URL": "",
  "X": -79.352625219,
  "Xmax": -79.342625219,
  "Xmin": -79.362625219,
  "Y": 43.751706789,
  "Ymax": 43.761706789,
  "Ymin": 43.741706789
},
"extent": {
  "xmax": -79.342625219,
  "xmin": -79.362625219,
  "ymax": 43.761706789,
  "ymin": 43.741706789
},
"location": {
  "x": -79.352625219,
  "y": 43.751706789
},
"score": 80,
"source": "Geospatial Competency Centre (GCC)"
}],
"spatialReference": {
  "latestWkid": 4326,
  "wkid": 4326
}
}
```

Sample Response #2

Using the magicKey =

"dHA9MCNsb2M9MTM1ODQ4MzMjbG5nPTMzI2huPTE4I2xicz0xMDk6NDUyNDU2MDM=", the detail attributes of "18 Dyana Dr, Lindsay, Kawartha Lakes, Ontario, K9V 5A3" can be retrieved.

https://map.toronto.ca/geoservices/rest/geocodeserver/findAddressCandidates?Single%20Line%20Input=&f=pjson&outFields=*&magicKey=dHA9MCNsb2M9MTM1ODQ4MzMjbG5nPTMzI2huPTE4I2xicz0xMDk6NDUyNDU2MDM=

```
{
  "candidates": [{
    "address": "18 Dyana Dr, Lindsay, Kawartha Lakes, Ontario, K9V 5A3",
    "attributes": {
      "AddBldg": "",
      "AddNum": "18",
      "AddNumFrom": "98",
      "AddNumTo": "2",
      "AddRange": "2-98",
      "Addr_type": "StreetAddress",
      "BldgName": "",
      "BldgType": "",
      "Block": "",
      "City": "Kawartha Lakes",
      "Country": "CAN",
      "DisplayX": -78.74072252188621,
      "DisplayY": 44.366127711248964,
      "Distance": 114638.19603127624,
      "District": "Lindsay",
      "ExInfo": "",
      "LangCode": "ENG",
      "LevelName": "",
      "LevelType": "",
      "Loc_name": "World",
      "LongLabel": "18 Dyana Dr, Lindsay, Kawartha Lakes, Ontario, K9V 5A3,
CAN",
      "Match_addr": "18 Dyana Dr, Lindsay, Kawartha Lakes, Ontario, K9V 5A3",
      "MetroArea": "",
      "Nbrhd": "",
      "Phone": "",
      "PlaceName": "",
      "Place_addr": "18 Dyana Dr, Lindsay, Kawartha Lakes, Ontario, K9V 5A3",
      "Postal": "K9V",
      "PostalExt": "5A3",
      "Rank": 20,
      "Region": "Ontario",
      "RegionAbbr": "ON",
      "Score": 100,
      "Sector": "",
      "ShortLabel": "18 Dyana Dr",
      "Side": "R",
      "StAddr": "18 Dyana Dr",
      "StDir": "",
      "StName": "Dyana",
    }
  ]
}
```

```
    "StPreDir": "",
    "StPreType": "",
    "StType": " Dr",
    "Status": "M",
    "SubAddr": "",
    "Subregion": "Kawartha Lakes",
    "Territory": "",
    "Type": "",
    "URL": "",
    "UnitName": "",
    "UnitType": "",
    "X": -78.74072252188621,
    "Xmax": -78.73972252188621,
    "Xmin": -78.74172252188622,
    "Y": 44.366127711248964,
    "Ymax": 44.36712771124896,
    "Ymin": 44.36512771124897,
    "Zone": ""
  },
  "extent": {
    "xmax": -78.73972252188621,
    "xmin": -78.74172252188622,
    "ymax": 44.36712771124896,
    "ymin": 44.36512771124897
  },
  "location": {
    "x": -78.74072252188621,
    "y": 44.366127711248964
  },
  "score": 100
}],
"spatialReference": {
  "latestWkid": 4326,
  "wkid": 4326
}
}
```

NOTES: this location is outside Toronto, data is retrieved from esri World Geocoder

2.3.2 findAddressCandidates by Single Line Input

Sample Response #1: search by input string of "18 dyas"

<https://map.toronto.ca/geoservices/rest/geocodeserver/findAddressCandidates?f=json&Single%20Line%20Input=18%20dyas>

```
{
  "candidates": [{
    "address": "18 Dyas Rd, North York, Ontario, M3B",
    "attributes": {},
    "extent": {
      "xmax": -79.342625219,
      "xmin": -79.362625219,
      "ymax": 43.761706789,
      "ymin": 43.741706789
    },
    "location": {
```

```
        "x": -79.352625219,  
        "y": 43.751706789  
    },  
    "score": 97,  
    "source": "Geospatial Competency Centre (GCC)"  
  }],  
  "spatialReference": {  
    "latestWkid": 4326,  
    "wkid": 4326  
  }  
}
```

Note: No outSrid has been given, the OUTX and OUTY are in WGS84 by default

Sample Response #2: search by input string of "18 dyas" with all fields

https://map.toronto.ca/geoservices/rest/geocodeserver/findAddressCandidates?f=pjson&Single%20Line%20Input=18%20dyas&outFields=*

```
{  
  "candidates": [{  
    "address": "18 Dyas Rd, North York, Ontario, M3B",  
    "attributes": {  
      "AddBldg": "",  
      "AddNum": "18",  
      "AddNumFrom": "ADDNUMFROM",  
      "AddNumTo": "ADDNUMTO",  
      "Addr_type": "PointAddress",  
      "City": "North York",  
      "Country": "CAN",  
      "DisplayX": -79.352625219,  
      "DisplayY": 43.751706789,  
      "Distance": 0,  
      "LangCode": "ENG",  
      "Loc_name": "CAN.PointAddress",  
      "Match_addr": "18 Dyas Rd, North York",  
      "Nbrhd": "",  
      "Phone": "",  
      "PlaceName": "",  
      "Place_addr": "",  
      "Postal": "M3B",  
      "PostalExt": "",  
      "Rank": "",  
      "Region": "Ontario",  
      "Score": 97,  
      "Side": "R",  
      "StAddr": "18 Dyas Rd",  
      "StDir": "",  
      "StName": "Dyas",  
      "StPreDir": "",  
      "StPreType": "",  
      "StType": "Rd",  
      "Subregion": "",  
      "Type": "",  
      "URL": "",  
      "X": -79.352625219,  
      "Xmax": -79.342625219,  
      "Y": 43.751706789,  
      "Ymax": 43.761706789  
    }  
  ]  
}
```



```
    "Xmin": -79.362625219,
    "Y": 43.751706789,
    "Ymax": 43.761706789,
    "Ymin": 43.741706789
  },
  "extent": {
    "xmax": -79.342625219,
    "xmin": -79.362625219,
    "ymax": 43.761706789,
    "ymin": 43.741706789
  },
  "location": {
    "x": -79.352625219,
    "y": 43.751706789
  },
  "score": 97,
  "source": "Geospatial Competency Centre (GCC)"
}],
"spatialReference": {
  "latestWkid": 4326,
  "wkid": 4326
}
}
```

Sample Response #3: search for intersection by an input string "finch / yong"

http://insidetoronto.ca/geoservices/rest/services/gccgeocoder/GeocodeServer/findAddressCandidates?Single%20Line%20Input=finch%20/%20yong&torontoOnly=true&f=pjson&outFields=*

```
{
  "candidates": [
    {
      "address": "Finch Ave W & Yonge St, North York,Ontario",
      "attributes": {
        "AddBldg": "",
        "AddNum": "",
        "AddNumFrom": "",
        "AddNumTo": "",
        "Addr_type": "StreetInt",
        "City": "",
        "Country": "CAN",
        "DisplayX": -79.415577699,
        "DisplayY": 43.779769082,
        "Distance": 0,
        "LangCode": "ENG",
        "Loc_name": "CAN.StreetAddress",
        "Match_addr": "Finch Ave W & Yonge St, North York",
        "Nbrhd": "",
        "Phone": "",
        "PlaceName": "",
        "Place_addr": "",
        "Postal": "",
        "PostalExt": "",
        "Rank": "",
        "Region": "",

```

```
    "Score": 84,
    "Side": "",
    "StAddr": "",
    "StDir": "",
    "StName": "",
    "StPreDir": "",
    "StPreType": "",
    "StType": "",
    "Subregion": "",
    "Type": "",
    "URL": "",
    "X": -79.415577699,
    "Xmax": -79.405577699,
    "Xmin": -79.425577699,
    "Y": 43.779769082,
    "Ymax": 43.789769082,
    "Ymin": 43.769769082
  },
  "extent": {
    "xmax": -79.405577699,
    "xmin": -79.425577699,
    "ymax": 43.789769082,
    "ymin": 43.769769082
  },
  "location": {
    "x": -79.415577699,
    "y": 43.779769082
  },
  "score": 84,
  "source": "Geospatial Competency Centre (GCC)"
},
{
  "address": "Finch Ave E & Yonge St, North York,Ontario",
  "attributes": {
    "AddBldg": "",
    "AddNum": "",
    "AddNumFrom": "",
    "AddNumTo": "",
    "Addr_type": "StreetInt",
    "City": "",
    "Country": "CAN",
    "DisplayX": -79.415577699,
    "DisplayY": 43.779769082,
    "Distance": 0,
    "LangCode": "ENG",
    "Loc_name": "CAN.StreetAddress",
    "Match_addr": "Finch Ave E & Yonge St, North York",
    "Nbrhd": "",
    "Phone": "",
    "PlaceName": "",
    "Place_addr": "",
    "Postal": "",
    "PostalExt": "",
    "Rank": "",
    "Region": "",
    "Score": 84,
    "Side": "",
    "StAddr": "",
```

```

        "StDir": "",
        "StName": "",
        "StPreDir": "",
        "StPreType": "",
        "StType": "",
        "Subregion": "",
        "Type": "",
        "URL": "",
        "X": -79.415577699,
        "Xmax": -79.405577699,
        "Xmin": -79.425577699,
        "Y": 43.779769082,
        "Ymax": 43.789769082,
        "Ymin": 43.769769082
    },
    "extent": {
        "xmax": -79.405577699,
        "xmin": -79.425577699,
        "ymax": 43.789769082,
        "ymin": 43.769769082
    },
    "location": {
        "x": -79.415577699,
        "y": 43.779769082
    },
    "score": 84,
    "source": "Geospatial Competency Centre (GCC)"
}
],
"spatialReference": {
    "latestWkid": 4326,
    "wkid": 4326
}
}

```

2.4 reverseGeocode Service (esri format)

<https://insidetoronto-map.toronto.ca/geoservices/rest/services/gccgeocoder/GeocodeServer/reverseGeocode>

Description: Find the nearest address or intersection by coordinates

Input Parameters

Parameter	Type	Required	Description
location	string	Yes	an location object, e.g. {"x":313639.277,"y":4845019.146,"spatialReference":{"wkid":2952}}
distance	integer	No	Distance / radius for proximity search, in meters Default is 100

outSR	integer	No	Output SRID. A Spatial Reference ID MTM NAD27 – 2019 Web Mercator – 3857 WSG84 – 4326 NAD83 CSRS – 2952 Default is WSG84 4326
featureTypes	string	No	Feature Types returned < PointAddress StreetInt > Default is PointAddress
returnRows	integer	No	Maximum number of rows returned Default is 100
torontoOnly	boolean	No	Toronto Only < true false > Default is false
useWorldCoverage	Boolean	No	Use world coverage for search
f	string	No	format < json pjson xml>

Sample Response #1: search for address from a given co-ordinate

<https://insideto-map.toronto.ca/geoservices/rest/services/gccgeocoder/GeocodeServer/reverseGeocode?location={%22x%22:-79.3841,%22y%22:43.6534,%22spatialReference%22:{%22wkid%22:4326}}&featureTypes=PointAddress&torontoOnly=false&f=pjson>

```
{
  "address": {
    "ADDNUM": "100",
    "ADDRESS": "100 Queen St W",
    "ADDR_TYPE": "PointAddress",
    "BLOCK": null,
    "CITY": "Toronto",
    "COUNTRYCODE": "CAN",
    "DISTRICT": "former Toronto",
    "LONGLABEL": "100 Queen St W, former Toronto, Toronto, Ontario, M5H, CAN",
    "MATCH_ADDR": "100 Queen St W, former Toronto, Toronto, Ontario, M5H",
    "METROAREA": "Greater Toronto Metro Area",
    "NEIGHBORHOOD": "Bay Street Corridor (76)",
    "PLACENAME": "City Hall - Nathan Phillips Square",
    "POSTAL": "M5H",
    "POSTALEXT": null,
    "REGION": "Ontario",
    "SECTOR": null,
    "SHORTLABEL": "100 Queen St W",
    "SUBREGION": "Toronto",
    "TERRITORY": null,
    "TYPE": null
  },
  "location": {
    "spatialReference": {
      "latestWkid": 4326,
```

```
        "wkid": 4326
      },
      "x": -79.383625059,
      "y": 43.65282214
    },
    "source": "Geospatial Competency Centre (GCC)"
  }
}
```

Sample Response #2: search for intersection from a given co-ordinate

[https://insidetoronto.ca/geoservices/rest/services/gccgeocoder/GeocodeServer/reverseGeocode?location={\"x\":-79.3841,\"y\":43.6534,\"spatialReference\":{\"wkid\":4326}}&featureTypes=StreetInt&torontoOnly=false&f=pjson](https://insidetoronto.ca/geoservices/rest/services/gccgeocoder/GeocodeServer/reverseGeocode?location={\)

```
{
  "address": {
    "ADDNUM": null,
    "ADDRESS": "Elizabeth St & Hagerman St",
    "ADDR_TYPE": "StreetInt",
    "BLOCK": null,
    "CITY": "Toronto",
    "COUNTRYCODE": "CAN",
    "DISTRICT": "former Toronto",
    "LOGLABEL": "Elizabeth St & Hagerman St, former Toronto, Toronto, Ontario, M5G, CAN",
    "MATCH_ADDR": "Elizabeth St & Hagerman St, former Toronto, Toronto, Ontario, M5G",
    "METROAREA": "Greater Toronto Metro Area",
    "NEIGHBORHOOD": "Bay Street Corridor (76)",
    "PLACENAME": null,
    "POSTAL": "M5G",
    "POSTALEXT": null,
    "REGION": "Ontario",
    "SECTOR": null,
    "SHORTLABEL": "Elizabeth St & Hagerman St",
    "SUBREGION": "Toronto",
    "TERRITORY": null,
    "TYPE": null
  },
  "location": {
    "spatialReference": {
      "latestWkid": 4326,
      "wkid": 4326
    },
    "x": -79.384602229,
    "y": 43.654054893
  },
  "source": "Geospatial Competency Centre (GCC)"
}
```

Appendix A

Full List of Area Type Codes (currency varies)

Area Type Code	Description
AD	Assessment Division
ARCH	Archaeological Area
ASA	Assessment Area
ASAR	Archaeological Sensitive Area (for internal use only)
ASD	Assessment Subdivision
BIA	Business Improvement Area
CC	Community Council
CENS	Census Tract
CITW	Toronto City Ward
CNBH	Census Neighbourhood
CONC	Concession Block
CPLN	Community Planning Boundary
CRGD	TMMS Crossing Guard Area
DARA	Dissemination Area
EPS	English Public School
ESS	English Separate School
FED	Federal Electoral District
FOCT	TMMS FO Commercial Tree
FORD	Forestry District
FORR	Forestry Region
FORV	Forestry Ravine District
FOWB	Forestry Operational Work
FPS	French Public School

FSA	Forward Sortation Areas (first 3 characters of postal code)
FSS	French Separate School
IBMD	IBMS District
IBMG	IBMS Grid
MUN	Former Municipality
NTWC	Night Time Waste Collection Schedule
OPR	SHAS TCHC Operation
PDIS	Police District
PDIV	Police Division
PZON	Police Zone
PDMB	Property Data Block
PDMS	Property Data Sub-block
PPA	Parking Permit Area
PR	Provincial Riding
PSW	Significant Wetland
PTPA	Permit Parking Area
RBL	Ravine Bylaw
RESW	Residential Waste Collection Schedule
SAM	Street Markings Area
SDIS	Social Services District
SIGD	Sign by-law District
SIGS	Sign by-law Special Area
SRA	Sidewalk Snow Removal
SSOF	Social Services Office Area
SWHW	Household Hazardous Waste
SWLA	Solid Waste Litter Area
SWMD	Solid Waste Management District

TD	TD Road Operations Maintenance
TFRA	Toronto Fire Run Area
TIA	TMMS Investigator Area
TOA	TMMS Operation Area
TPLD	TO Public Library Districts
TPLN	TO Public Library Neighbourhood
TRCA	TO Regional Conservation Authority Boundary
UPDD	Urban Planning & Development District
VSD	Voting Subdivision
WD03	Toronto Municipal Electoral Ward (44)
WD18	Toronto Municipal Electoral Ward (25)
WGMD	Solid Waste White Goods
WTPZ	Water Pressure Zone